

Al-Aqsa University
Faculty of Engineering
Intelligence Systems &
Computer Engineering Dept.



جامعة الأقصى
كلية الهندسة
قسم هندسة النظم الذكية و الحاسوب

Smartphones Programming (ENGS4301)

Lab 1: Introduction to Dart

Presented to: Eng. Mohammed Iyad Al-Laham

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Student number: 2320222601

[Link to Github](#)

Semester: Second Semester 2025/2026

Question 1: Write a Dart program that prints information:

Code

```
Lect-01-Introducton-to-dart > Q1-print.dart > ...
1 // Question 1 : Write a Dart program that prints information
2
Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X
3 void main() {
4     print("Student Name: Samar M. Balousha ");
5     print("Student Age: 21 years old");
6     print("Student Major: Intelligence Systems and Computer Engineering ");
7
8 }
```

Result

```
PS D:\Level4\Smartphone-Programming\Smartphone-Programming> & 'D:\Level4\Smartphone-Programming\dart-sdk-windows-x64-release\dart-sdk\bin\dart.exe' '--no-serve-devtools' '--enable-asserts' 'D:\Level4\Smartphone-Programming\Smartphone-Programming\Lect-01-Introducton-to-dart\Q1-print.dart'
Student Name: Samar M. Balousha
Student Age: 21 years old
Student Major: Intelligence Systems and Computer Engineering
```

Question 2: Arithmetic operations (sum, subtract, multiply, division, reminder)

Code

```
Lect-01-Introducton-to-dart > Q2-Arithmetic-Operation.dart > main
1 // Question 2 : Arithmetic Operation
2 // Define 2 numbers:
Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X
3 void main() {
4     int number1 = 15;
5     int number2 = 12;
6
7     // operations
8     int sum = number1 + number2 ;
9     int subtraction= number1 - number2 ;
10    int multiply = number1 * number2 ;
11    double division = number1 / number2 ;
12    int remainder = number1 % number2;
13
14    //print results
15    print("The sum of $number1 and $number2 is: $sum");
16    print("The subtraction of $number1 and $number2 is: $subtraction");
17    print("The multiplication of $number1 and $number2 is: $multiply");
18    print("The division of $number1 and $number2 is: $division");
19    print("The remainder of $number1 and $number2 is: $remainder");
20 }
```

Result

```
--no-serve-devtools --enable-asserts D:\Level14\Smartphone-Programming\Smartphone-Pr  
The sum of 15 and 12 is: 27  
The subtraction of 15 and 12 is: 3  
The multiplication of 15 and 12 is: 180  
The division of 15 and 12 is: 1.25  
The remainder of 15 and 12 is: 3
```

Question 3: Evaluating Complex Expressions

Code

```
Lect-01-Introduction-to-dart > Q3-Complex-Expressions.dart > ...  
1 // Question 3 : Evaluating Complex Expressions  
  Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X  
2 void main() {  
3   int expression1 = -5+8*6;  
4   print('Expression 1: -5 + 8 * 6 = $expression1');  
5  
6   int expression2 = (55+1)%9;  
7   print('Expression 2: (55 + 1) % 9 = $expression2');  
8  
9   double expression3 = 20 + -3*5 / 8;  
10  print('Expression 3: 20 + -3 * 5 / 8 = $expression3');  
11  
12  double expression4 = 5 + 15 / 3 * 2 - 8 % 3;  
13  print('Expression 4: 5 + 15 / 3 * 2 - 8 % 3 = $expression4');  
14  
15 }  
16
```

Result

```
g:\Lect-01-Introduction-to-dart\Q3-Complex-Expressions\dart  
Expression 1: -5 + 8 * 6 = 43  
Expression 2: (55 + 1) % 9 = 2  
Expression 3: 20 + -3 * 5 / 8 = 18.125  
Expression 4: 5 + 15 / 3 * 2 - 8 % 3 = 13.0
```

Question 4: Print information

Code

```
Lect-01-Introducion-to-dart > Q4-Profile-data.dart > main
1 // Question 4 : Write a Dart program to print your complete personal informati
Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X
2 void main(){
3 // variables
4 String name = 'Samar M. Balousha';
5 String university = 'Al-Aqsa University';
6 String universityNumber = '2320222601';
7 String college = 'Engineering';
8 String address = 'Gaza';
9 String maritalStatus = 'Single';
10
11 // print Profile data
12 print('' Profile Data Info :
13 Student Name: $name
14 University: $university
15 University Number: $universityNumber
16 College: $college
17 Address: $address
18 Marital Status: $maritalStatus'');
19
20 }
```

Result

g\Lect-01-Introducion-to-dart\Q4-Profile-data.dart'

```
Profile Data Info :
Student Name: Samar M. Balousha
University: Al-Aqsa University
University Number: 2320222601
College: Engineering
Address: Gaza
Marital Status: Single
```

D:\Level1\Smartphone-Programming\Smartphone-Programming\

Question 5: Draw a shape

Code

Lect-01-Introduction-to-dart >  Q5-Shape.dart >  main

```
1 // Question 5 : Shape
  Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X
2 void main() {
3   print ("--- The Shape ---");
4   print(''
5     + "'''''''''+
6     [|  o o  |]
7     |   ^   |
8     |  ' - ' |
9     +-----+
10    '');
11 }
```

Result

```
--no-serve-devtools --enable-asserts
--- The Shape ---
+'''''''''+
[|  o o  |]
|   ^   |
|  ' - ' |
+-----+
```

Question 6: Area of rectangle

Code

```
Lect-01-Introducion-to-dart > Q6-rectangle-area.dart > main
1 // Question 6 : area of a rectangle
2 import 'dart:io';
3
4 void main() {
5     stdout.write('Enter the length of the rectangle: ');
6     double length = double.parse(stdin.readLineSync()!);
7
8     stdout.write('Enter the width of the rectangle: ');
9     double width = double.parse(stdin.readLineSync()!);
10
11     double area = length * width;
12
13     print('The area of the rectangle is: $area');
14 }
15
```

Result

```
g\lect-01-introducion-to-dart\Q6-rectangle-area.dart
Enter the length of the rectangle: 15.2
Enter the width of the rectangle: 3.1
The area of the rectangle is: 47.12
```

Question 7: Area of a circle

Code

```
Lect-01-Introducion-to-dart > Q7-Circle-area.dart > main
1 // question 7 : get the circle area from user inout
2 import 'dart:io';
3
4 void main () {
5     // constant
6     const double pi = 3.14;
7     stdout.write('Enter the radius : ');
8     double radius = double.parse(stdin.readLineSync()!);
9     double area = pi * radius * radius;
10    print('The area of the circle is: $area');
11 }
```

Result

```
g\Lect-01-Introdution-to-dart\Q7-Circle-area.dart'  
Enter the radius : 7  
The area of the circle is: 153.86
```

Question 8: Shekel to Dollar

Code

```
Lect-01-Introdution-to-dart > Q8-Shekel-to-dollar.dart > main  
1 // Question 8:m Shekel to dollar converter  
2 import 'dart:io';  
   Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X  
3 void main () {  
4   // constant  
5   const double exchange = 3.25 ; // 1 dollar = 3.25 shekels  
6   stdout.write('Enter the amount in Shekels: ... ');  
7   double shekel = double.parse(stdin.readLineSync());  
8   double dollar = shekel / exchange;  
9   print('The amount in dollars is: $dollar');  
10 }
```

Result

```
g\Lect-01-Introdution-to-dart\Q8-Shekel-to-dollar.dart'  
Enter the amount in Shekels: ... 100  
The amount in dollars is: 30.76923076923077
```

Question 9: Temperature Converter

Code

```
Lect-01-Introdution-to-dart > Q9-Temperature-converter.dart > main  
1 // Question 9 : Temperature Converter Fahrenheit to Celsius  
2 import 'dart:io';  
   Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X  
3 void main() {  
4   stdout.write('Enter the temperature in Fahrenheit: ... ');  
5   double fahrenheit = double.parse(stdin.readLineSync());  
6   double celsius = 5 * (fahrenheit - 32) / 9;  
7   print('The temperature in Celsius is: $celsius');  
8 }
```

Result

```
g\Lect-01-Introduction-to-dart\Q9-Temperature-converter.dart'  
Enter the temperature in Fahrenheit: ... 90  
The temperature in Celsius is: 32.22222222222222  
PS D:\Level4\Smartphone-Programming\Smartphone-Programming>
```

Question 10: Length Converter

Code

```
Lect-01-Introduction-to-dart > Q10-Length-Converter.dart > main  
1 // Question 10 : Length convertor ( inch to meter)  
2 import 'dart:io';  
   Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X  
3 void main() {  
4     stdout.write('Enter the length in inches: ... ');  
5     double inch = double.parse(stdin.readLineSync());  
6     double meter = inch * 0.0254;  
7     print('The length in meters is: $meter');  
8 }
```

Result

```
g\Lect-01-Introduction-to-dart\Q10-Length-Converter.dart'  
Enter the length in inches: ... 5.3  
The length in meters is: 0.13462
```


Question 11: Summing Digits of a Number

Code

```
Lect-01-Introducton-to-dart > Q11-Sum-of-digits.dart > main
1 // Question 11: Summing teh digits of a number 0 and 1000
2 import 'dart:io';
   Run | Debug | Windsurf: Refactor | Explain | Generate Function Comment | X
3 void main() {
4     stdout.write('Enter a number between 0 and 1000: ... ');
5     int number = int.parse(stdin.readLineSync()!);
6
7     if (number < 0 || number > 1000) {
8         print('Please enter a valid number between 0 and 1000.');
```

```

9         return;}
10    print("-----");
11
12    int sum = 0;
13    int originalNumber = number;
14    print('The process of summing the digits of $number:');
```

```

15    while (originalNumber > 0) {
16        // add the last digit to the sum
17        sum += originalNumber % 10;
18        // remove the last digit
19        originalNumber ~/= 10;
20        print('Current sum: $sum, Remaining number: $originalNumber');
```

```

21    }
22    print('The sum of the digits of $number is: $sum');
```

```

23 }
```

Result

```
Enter a number between 0 and 1000: ... 982
-----

The process of summing the digits of 982:
Current sum: 2, Remaining number: 98
Current sum: 10, Remaining number: 9
Current sum: 19, Remaining number: 0
The sum of the digits of 982 is: 19
```

Question 12: Time Converter

Code

Lect-01-Introducton-to-dart >  Q12-Time-converter.dart >  main

```
1 // Question 12: Time Converter from minutes to days and years
2 import 'dart:io';
3 void main(){
4   stdout.write('Enter the time in minutes : __');
5   int minutes = int.parse(stdin.readLineSync()!);
6   int days = minutes ~/ (60 * 24);
7   int years = days ~/ 365;
8   int remainderDays = days % 365;
9   print('$minutes minutes is approximately $years years and $remainderDays days');
10 }
```

Result

```
ramming\dart-sdk-windows-x64-release\dart-sdk\bin\dart.exe' --no-serve-devtool
g\Lect-01-Introducton-to-dart\Q12-Time-converter.dart'
Enter the time in minutes : __852654
852654 minutes is approximately 1 years and 227 days
```